sEMG Muscle Activation Report

Participant P9 · Gesture_1 · April 17, 2026

Overall Activation	Very High
Temporal Profile	Sustained
Consistency	Consistent
Dominant Sensors	S2
Weakest Sensors	S8
Assessment Confidence	0.95

Clinical Summary: The session demonstrates a sustained, flexor-dominant contraction pattern. High co-contraction (0.55) suggests a power-grip style activation. S2 and S3 are the primary contributors to force production. S1 exhibits inconsistent activation compared to other sensors.

Key Findings:
- High co-contraction ratio
- Flexor-dominant activation pattern
- S1 intermittent profile

Session Overview

Session Duration	1203.4 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.015197 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.012799	0.410	101.2	89.3	1657	High	Intermittent
S2	0.031093	0.397	106.0	99.2	6204	Very High	Sustained
S3	0.020936	0.144	101.4	91.1	4251	Very High	Sustained
S4	0.015532	0.129	79.1	64.5	1251	Very High	Ramp-up
S5	0.012339	0.190	125.3	111.9	1753	High	Sustained
S6	0.007643	0.172	130.4	113.1	827	Moderate	Sustained
S7	0.015205	0.119	114.2	104.1	1983	Very High	Sustained
S8	0.006031	0.166	111.0	91.6	126	Moderate	Ramp-up

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S1: High variability in intermittent activation

S2: Primary driver of sustained contraction

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S2	Primary contributor
Weakest Sensor	S8	Lowest activity
Dominant-to-Weakest Ratio	5.16×	Activation disparity
Concentration Index	0.161	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.994	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.257	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.553	Agonist/antagonist balance
Flexor/Extensor Balance	1.98	>1 = flexor-dominant

Overall Temporal Profile	Sustained	Session-wide activation trend

Co-contraction: Detected — Agonist and antagonist muscles activating simultaneously (normal for power grip).

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows significantly higher activation levels compared to extensor group (S5-S8), with a flexor-extensor balance ratio of 1.98.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S2, S3

Evidence: High WAMP values observed in S2 (6204.4) and S3 (4250.6) indicate significant motor unit recruitment during sustained effort.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.